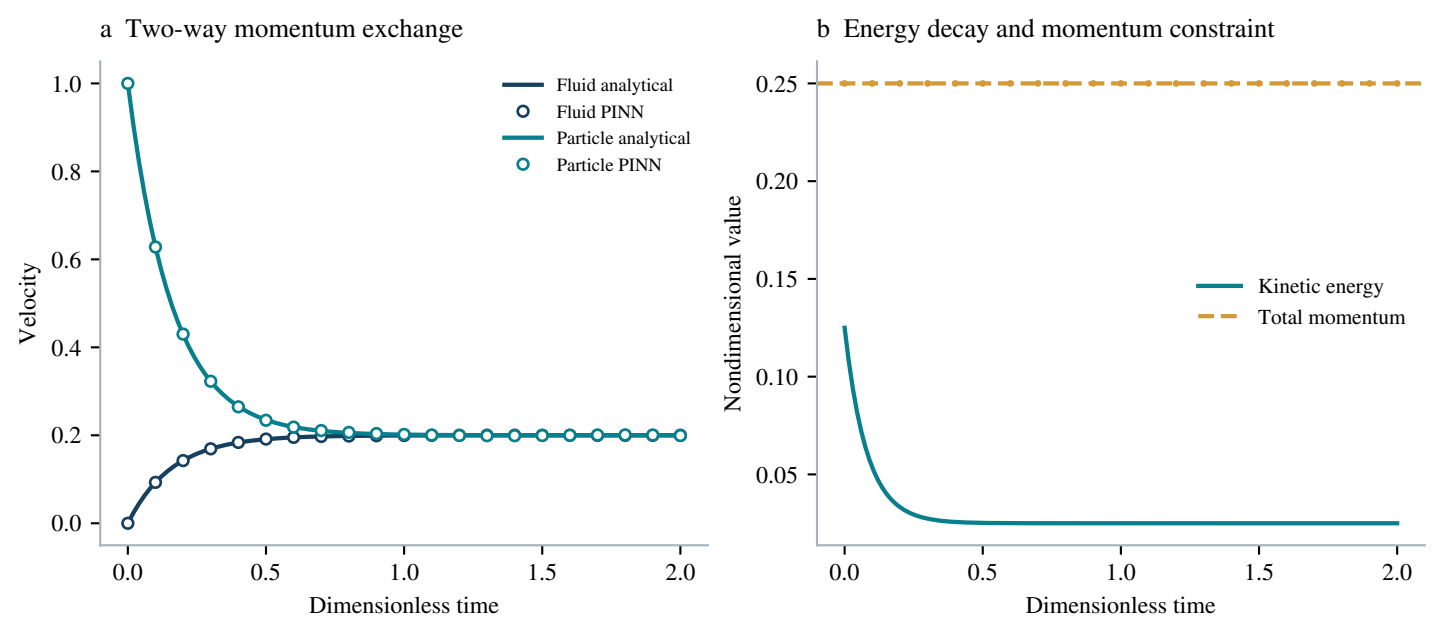
a Two-way momentum exchange
Fluid analytical
Fluid PINN
Particle analytical
Particle PINN
Velocity
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.5
1.0
1.5
2.0
Dimensionless time
b Energy decay and momentum constraint
Kinetic energy
Total momentum
Nondimensional value
0.25
0.20
0.15
0.10
0.05
0.0
0.5
1.0
1.5
2.0
Dimensionless time